

Static Gradient Attribution Underperforms a Density-Matched Random Mask on Loss-Based Forgetting Within LoRA’s B-Matrix

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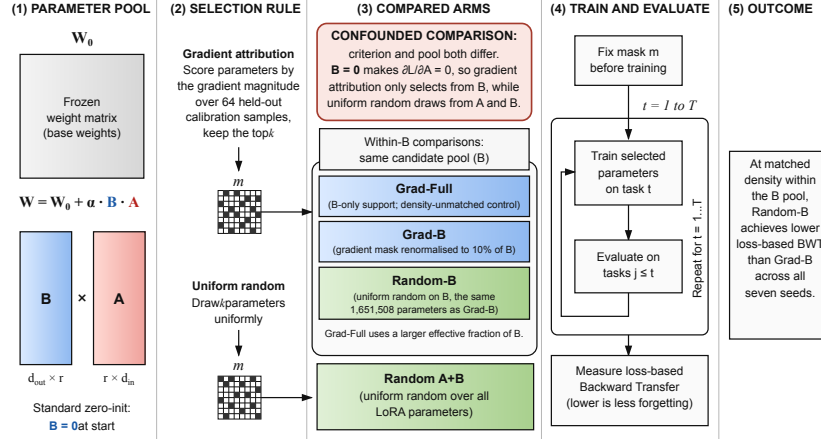
Abstract

Parameter-efficient continual learning updates a restricted subset of adapter parameters to incorporate sequential tasks while limiting catastrophic forgetting. Gradient attribution is a common rule for selecting that subset, but it requires an additional backward pass over calibration data. We identify a parameter-pool confound in comparisons between initialization-time gradient masks and unrestricted random masks under standard zero-initialized LoRA ($B = 0$): the gradient with respect to A is identically zero ($\partial\mathcal{L}/\partial A = 0$), so gradient masks select strictly from the B matrix whereas unrestricted random masks draw from both A and B . Such a comparison changes both the selection rule and the candidate parameter pool. Isolating the selection criterion by restricting both approaches to the same parameter pool, we find a negative result: a uniform random mask consistently achieves lower loss-based backward transfer than gradient attribution across seven seeds on Qwen2.5-3B. The effect is concentrated on the medical task, which contributes 80% of the aggregate; code and math move in the same direction, and legal shows no reliable difference. This advantage is consistent with a stability–plasticity trade-off under the tested task sequence, in which random masking slightly reduces single-task fitting. On the two discrete-option tasks (Medical and Legal), the mean final-accuracy difference is not statistically distinguishable from zero. The 95% interval extends to 4.80 percentage points in the gradient mask’s favour, so the pre-specified 4-point non-inferiority criterion is not met. On loss-based forgetting, Random-B consistently outperforms the density-matched gradient mask across seven seeds; on accuracy, the evidence remains inconclusive at $n = 7$. The attribution pass is therefore not justified by the loss-based forgetting evidence, while its accuracy consequences require a better-powered evaluation.

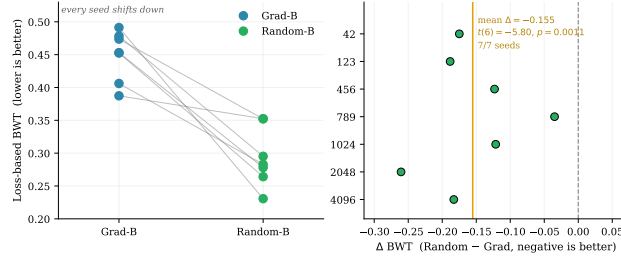
1 Introduction

Parameter-efficient adapters such as LoRA (Hu et al., 2021; Dettmers et al., 2023) make it possible to incorporate a sequence of tasks while updating only a small fraction of a language model. Sequential updates can nevertheless overwrite capabilities acquired on earlier tasks, the classical problem of catastrophic forgetting (McCloskey & Cohen, 1989; French, 1999). Continual learning for LoRA adapters updates only a chosen subset of parameters, and gradient attribution is the common rule for choosing it: a backward pass over held-out calibration data scores each parameter, and the highest-scoring ones are updated. Choosing the subset uniformly at random costs nothing instead, so whether the attribution pass earns its cost has a direct engineering answer.

Standard zero-initialized LoRA introduces a potential confound. With $B = 0$ at initialization, $\partial\mathcal{L}/\partial A = 0$, so gradient attribution assigns zero importance to every `lora_A` parameter and a gradient mask selects only from `lora_B`, while an unrestricted random baseline draws from both factors together. Any such comparison mixes the selection criterion with the parameter pool. Attribution-Guided CL (Liu et al., 2026) is the closest prior use of gradient



(a) Method and comparison.



(b) Key effect (D1, seven seeds).

Figure 1: The density-matched comparison. (a) Both arms draw 10% of the rank-16 `lora_B` pool, so they train exactly the same 1.6M parameters and differ only in how those coordinates are chosen; because $B = 0$ at initialization, the gradient score is zero for every `lora_A` coordinate, and any baseline that draws from both pools changes the pool as well as the criterion. (b) Under that matched density the random mask achieves lower loss-based BWT than the gradient mask in all seven seeds.

attribution as a continual-learning mask for LLMs; because its published description does not establish all initialization, attribution-timing, and mask-scope details needed to determine whether the same pool mismatch applies, we treat it as motivation rather than as an empirical instance of the confound.

We separate the two by restricting both arms to the same pool and parameter count. On Qwen2.5-3B with rank-16 LoRA over a five-task sequence we compare a uniformly random mask on `lora_B` (Random-B) against a gradient mask renormalised to the same 10% of `lora_B` (Grad-B), so both train exactly the same parameters and differ only in how those parameters are chosen. Across seven seeds the random mask achieves the lower loss-based backward transfer, and the gap holds in every seed (Table 2, Figure 1). The aggregate gap is concentrated on the medical task, with code and math moving in the same direction and legal showing no reliable difference, a pattern consistent with a stability-plasticity trade-off: the random mask fits each task less well on its own diagonal yet retains earlier tasks better. Above all, this work makes two key contributions:

Pool confound. Under zero-init LoRA, gradient masks select only from `lora_B`; an unrestricted random baseline therefore changes both the rule and the pool.

Density-matched evidence. Random-B has lower loss-based BWT in all seven seeds, but accuracy remains inconclusive (CI upper bound +4.80 pp; 4 pp non-inferiority unmet). On loss, attribution is not justified; removing it eliminates 64 calibration steps and their data requirement.

Table 1: Notation. Indices run over tasks $t = 1, \dots, T$ with $T = 5$, and over prior tasks $j \in \mathcal{P} = \{1, \dots, T - 1\}$.

Symbol	Meaning
f_θ, W_0	base model and its frozen pretrained weight matrix
B, A	low-rank LoRA factors, $B \in \mathbb{R}^{d_{\text{out}} \times r}$, $A \in \mathbb{R}^{r \times d_{\text{in}}}$
r, α	LoRA rank ($r = 16$) and scaling ($\alpha = 32$)
W	adapted weight, $W = W_0 + \alpha BA$
t, T	task index and number of tasks ($T = 5$)
$\mathcal{D}_t$	training split of task t
$\mathcal{D}$	held-out calibration set (64 Alpaca samples)
m	binary mask over the trainable coordinates, $m \in \{0, 1\}^{ A + B }$
k	selection budget: number of coordinates with $m = 1$
g_i	gradient-attribution score of coordinate i (§3.3)
$\mathcal{L}_j^{(\tau)}$	evaluation loss on task j after training through task τ
$\mathcal{P}$	prior-task set $\{1, \dots, T - 1\}$
BWT	loss-based backward transfer, Eq. equation 2

2 Background

Continual learning (CL) and Catastrophic Forgetting: (McCloskey & Cohen, 1989; French, 1999). A dominant strategy identifies important parameters and protects them during later tasks: EWC (Kirkpatrick et al., 2017) uses the diagonal Fisher information as a quadratic penalty, SI (Zenke et al., 2017) accumulates path-integral importance, and gradient-based selective-update methods score parameters from calibration gradients, an idea since extended to LLMs and parameter-efficient tuning (Qiao et al., 2025; Wang et al., 2025). Its incremental cost is an additional forward-backward pass over a calibration set, together with the associated compute time and calibration-data requirement. In our setup the attribution pass does not raise peak memory beyond the ordinary training backward pass (Appendix B).

Random Subspaces and Intrinsic Dimensionality: The alternative view is that the identity of the selected parameters matters less than how many are selected. The lottery ticket hypothesis (Frankle & Carbin, 2019) and supermasks (Zhou et al., 2019; Ramanujan et al., 2020) show random subnetworks in large networks can perform strongly, Aghajanyan et al. (2021) and Li et al. (2018) show that fine-tuning occupies a low-dimensional subspace, and DARE (Yu et al., 2024) shows that most delta parameters can be dropped. The same theme appears in pruning, where Liu et al. (2019) show pruned architectures can be retrained from scratch and Blalock et al. (2020) find that method choice matters less than the sparsity budget. LoRA already constrains updates to a low-rank subspace and masking restricts further within it, so the question is whether attribution beats a random restriction of the same size.

3 Methodology: Masking Mechanisms

3.1 Problem Formulation

We consider the standard sequential fine-tuning protocol for continual learning (French, 1999). A base model f_θ is adapted to task t on its training split $\mathcal{D}_t$ and then evaluated on all tasks $j \leq t$. Tasks arrive in a fixed order indexed by $t = 1, \dots, T$, with $T = 5$ throughout this paper.

We use LoRA adapters, which add a low-rank update $\Delta W = BA$ to a frozen pretrained weight matrix W_0 , where $B \in \mathbb{R}^{d_{\text{out}} \times r}$, $A \in \mathbb{R}^{r \times d_{\text{in}}}$, and $r \ll \min(d_{\text{out}}, d_{\text{in}})$:

$$W = W_0 + \alpha BA. \quad (1)$$

Table 1 summarizes the notation used throughout. A binary mask $m \in \{0, 1\}^{|A|+|B|}$ restricts optimization to coordinates for which $m_i = 1$. The mask is constructed once from

the calibration set before the first task and remains fixed throughout the task sequence (Algorithm 1).

We compare two coordinate-selection rules: gradient attribution, which ranks coordinates by their calibration-set gradient magnitude, and uniform random selection. For a fixed candidate parameter pool, both methods activate the same budget k ; they therefore differ only in which coordinates are selected.

We measure forgetting using loss-based backward transfer:

$$\text{BWT} = \frac{1}{|\mathcal{P}|} \sum_{j \in \mathcal{P}} \left(\mathcal{L}_j^{(T)} - \mathcal{L}_j^{(j)} \right), \quad (2)$$

where $\mathcal{P} = \{1, \dots, T-1\}$. This is the mean increase in evaluation loss on prior tasks after completing the full sequence; lower values indicate less forgetting.

3.2 The $B = 0$ Gradient-Attribution Confound

Standard LoRA initializes $B = 0$. For

$$W = W_0 + \alpha BA, \quad (3)$$

the gradient with respect to A is

$$\frac{\partial \mathcal{L}}{\partial A} = \alpha B^\top \frac{\partial \mathcal{L}}{\partial W}. \quad (4)$$

Hence, at initialization,

$$B = 0 \implies \frac{\partial \mathcal{L}}{\partial A} = 0. \quad (5)$$

Gradient attribution therefore assigns zero importance to every lora_A coordinate and selects exclusively from lora_B. In contrast, an unrestricted random mask samples from the full LoRA parameter pool, $\text{lora_A} \cup \text{lora_B}$. A direct comparison between these masks consequently changes two factors at once: the selection criterion and the candidate parameter pool.

To isolate the selection criterion, the candidate pool must therefore be matched. Restricting gradient attribution explicitly to lora_B is a null operation under zero initialization, whereas restricting the random baseline to lora_B removes the pool mismatch. Our primary comparison accordingly evaluates gradient and random selection within the same lora_B pool; Section 5.2 additionally matches their selection density.

This confound is specific to standard zero-initialized LoRA. For example, PiSSA (Meng et al., 2024) initializes both A and B from principal singular values, so $B \neq 0$ and Equation 4 does not force the gradient with respect to A to vanish. LoRA+ (Hayou et al., 2024) addresses related asymmetric optimization dynamics between the two LoRA factors. Whether gradient attribution is beneficial under non-zero initialization remains an open question.

3.3 Density-Matched Mask Construction

Let $\mathcal{S}$ denote the candidate set of LoRA coordinates from which a mask is selected. For gradient attribution, each coordinate $i \in \mathcal{S}$ is scored by its mean absolute gradient over the held-out calibration set $\mathcal{D}$:

$$g_i = \mathbb{E}_{x \sim \mathcal{D}} \left[\left| \frac{\partial \mathcal{L}(x)}{\partial w_i} \right| \right]. \quad (6)$$

Given a selection budget k , the binary mask activates the k highest-scoring coordinates:

$$m_i = \mathbf{1}[i \in \text{TopK}(\{g_j\}_{j \in \mathcal{S}}, k)]. \quad (7)$$

For uniform random selection, we instead draw independent scores

$$u_i \sim \text{Uniform}(0, 1), \quad i \in \mathcal{S}, \quad (8)$$

and apply the same top- k operator in Equation 7 with u_i in place of g_i . Thus, once the candidate pool $\mathcal{S}$ and budget k are fixed, the two methods differ only in the rule used to rank coordinates.

Density-matched comparison. Following the pool confound established in §3.2, our primary comparison restricts both selection rules to the `lora_B` coordinate pool, denoted by $\mathcal{B}$. Both headline arms activate exactly 10% of this pool:

$$k_{D1} = 1,651,508. \quad (9)$$

Grad-B applies Equation 6 within $\mathcal{B}$ and selects the top k_{D1} coordinates, whereas Random-B selects the same number of coordinates uniformly at random from $\mathcal{B}$. The two arms therefore have the same candidate pool and the same number of trainable coordinates; the selection criterion is the only varying factor.

Reference and control arms. For comparison with the conventional design, Grad-Full applies gradient attribution with the full LoRA coordinate set $\mathcal{A} \cup \mathcal{B}$ as its nominal candidate pool and uses a 10% full-pool budget of 2,993,357 coordinates. Under standard zero initialization, its selected support nevertheless lies entirely within $\mathcal{B}$ by §3.2; because its budget is larger than k_{D1} , Grad-Full is pool-matched to Random-B in effective support but not density-matched. Random A+B draws the same 10% full-pool budget uniformly from $\mathcal{A} \cup \mathcal{B}$ and therefore constitutes the conventional pool-unmatched random baseline.

We additionally use two auxiliary controls where relevant. No mask is standard LoRA with all adapter coordinates updated. Uniform-B uses the same `lora_B` budget as the matched comparison while distributing selections uniformly across LoRA modules; it is used only as a module-concentration control in §6.2.

Algorithm 1 summarizes mask construction and sequential training.

4 Experimental Setup

4.1 Models, Benchmarks & Training Details

Qwen2.5-3B (Team, 2024) in 4-bit NF4 (bitsandbytes), LoRA rank 16 ($\alpha=32$), AdamW (lr 2×10^{-4} , warmup 0.03), 1 epoch, 500 train samples per task, batch size 1, max length 512; calibration uses 64 Alpaca samples. All methods operate on 29.9M trainable LoRA parameters (0.96% of 3.1B); the full-pool top-10% selects 2,993,357, falling entirely within `lora_B` ($B = 0$) and activating $\sim 20\%$ of B, against 10% for Random-B. The comparison is matched on pool but not density; §5.2 equalizes it.

Five domains are used in a fixed order: Code (sahil2801/CodeAlpaca-20k), Medical (GBaker/MedQA-USMLE-4-options), Legal (lex_glue, casehold subset), Math (openai/gsm8k), Creative (ChaoticNeutrals/Creative_Writing-ShareGPT, 6.6K instruction-response pairs). Calibration: tatsu-lab/alpaca (64 samples); 100 eval samples per task.

4.2 Evaluation Metrics

We use Loss-based BWT in preference to accuracy-based BWT because loss is continuous and more sensitive to the small differences between masking strategies. All reported forgetting metrics are loss-based, reported as mean $\pm$ std over seven seeds (42, 123, 456, 789, 1024, 2048, 4096); part of the runs used a second GPU. The accuracy side of the same comparison, together with the option log-likelihood scorer and its scope, is reported separately in §5.4.

5 Experiments and Main Results

5.1 Confounded and Density-Matched Comparisons

A first comparison follows the usual design, in which each method draws from its own pool. Under that design gradient attribution shows no advantage: its loss-based BWT is 0.437 ± 0.061 against 0.428 ± 0.019 for a shared random mask over five seeds, a seed-dependent difference that is not significant (Table 5, Appendix D). Both methods still reduce forgetting relative to an unmasked adapter, but the comparison is uninformative about the selection rule, for the reason set out in §3.2: gradient attribution selects only `lora_B` while the random

baseline draws from both pools, so the design varies pool and criterion together. The result is reported for completeness and is distinct from the seven-seed comparison that follows, the two coming from different sessions whose absolute levels are not comparable (Appendix F).

Holding the pool fixed changes the picture. We introduce a Random-B control drawn from `lora_B` alone, matching the gradient mask’s pool, and compare it against a gradient mask restricted to the same pool. Each seed is an independent session, so absolute BWT values drift between runs (Appendix F) and only within-session paired observations are used. This comparison is matched on pool but not on density, since the gradient mask activates nearly twice as many `lora_B` coordinates as Random-B. Even so, the random mask achieves the lower loss-based BWT in all seven seeds, with a mean difference of -0.072 that survives dropping the largest-effect seed (two-sided paired $t(5) = -3.62$, $p = 0.015$; Figure 2, Appendix E). This value is the direct recomputation on the six retained paired contrasts and corrects the $p = 0.029$ value reported in an earlier draft. That the direction is unchanged when the pool is fixed is the first indication that the criterion, rather than the pool, is what the earlier comparison was measuring.

5.2 Density-Matched Control

Pool and criterion are separated cleanly only when density is matched as well. The comparison above leaves the gradient arm activating about 20% of the pool against 10% for Random-B, so the two arms still differ in how many coordinates they train. D1 removes that difference, with both arms drawing exactly 10% of `lora_B` and only the ranking rule differing. At matched density the random mask again achieves the lower loss-based BWT in every seed, by a larger margin than in the density-unmatched comparison (Table 2). Under a common [Grad-B – Random-B] convention, the mean contrast is $+0.155$ on the V100 stack and $+0.152$ on the A800 stack. Thus the contrast reproduces at a similar magnitude in a second hardware/software environment; in a separate three-seed sensitivity check, reversing the execution order of the two arms does not reverse the Random-B–Grad-B contrast (Appendix F).

5.3 Task-Level Decomposition

The aggregate gain is not spread evenly across the five tasks, and where it concentrates bears on how the effect should be read. Medical carries about four fifths of it, while code and math contribute the remainder in the same direction; on legal the arms are indistinguishable. Two explanations can be assessed from the same data. First, the gradient arm’s loss on each task’s own diagonal is lower than the random arm’s in every seed and task, yet it subsequently forgets more, so the two arms differ in retention rather than initial learning capacity. Endpoint reconstruction further shows that the contrast is not explained merely by a lower starting point (Appendix R). By contrast, restoring the gradient arm’s full budget

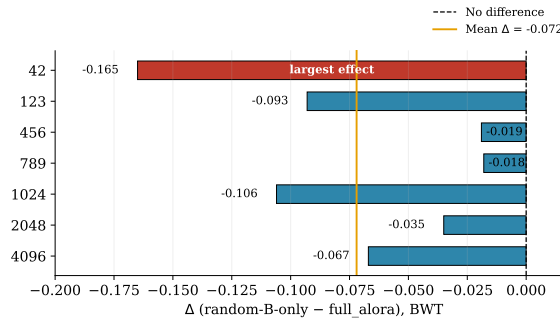


Figure 2: Per-seed within-`lora_B` differences ($\Delta = \text{Random-B} - \text{Grad-Full}$, loss-based BWT, 7 seeds). All seven are negative (Random-B forgets less), mean $\Delta = -0.072$ (orange line); seed 42 is the largest effect (-0.165). Leave-one-out excluding seed 42 remains significant (paired $t(5)$, $p = 0.015$).

Table 2: Headline result (D1, density-matched). Both arms select the same 1,651,508 parameters (10% of lora_B); only the criterion differs. Per-seed loss-based BWT and $\Delta = \text{Random-B} - \text{Grad-B}$ (negative = random-B forgets less), seven sessions, one seed each. The last column is the density-unmatched contrast from the same sessions, that is, Random-B against the unrestricted gradient mask ($\Delta = \text{Random-B} - \text{Grad-Full}$) rather than against Grad-B.

Seed	Random-B BWT ↓	Gradient-B BWT ↓	Δ (matched)	Δ (unmatched)
42	0.278	0.453	−0.175	−0.220
123	0.264	0.453	−0.188	−0.119
456	0.283	0.406	−0.123	−0.131
789	0.352	0.387	−0.035	−0.074
1024	0.352	0.474	−0.121	−0.109
2048	0.231	0.491	−0.261	−0.187
4096	0.295	0.478	−0.183	−0.166
Mean	0.294	0.449	−0.155	−0.144

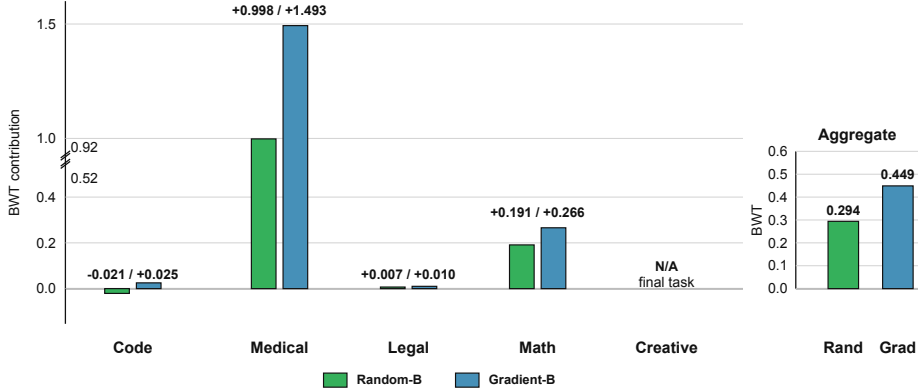


Figure 3: The density-matched D1 per-task decomposition, averaged over seven seeds. For each prior task, the bars show the loss-based BWT contribution of Random-B and Grad-B, with the corresponding values reported above each pair. Because the task-level contributions span roughly two orders of magnitude, the left panel uses a broken y -axis: the slashes mark the omitted interval from 0.52 to 0.92, and the labels above the Medical bars report their full values (+0.998 and +1.493) rather than truncated heights. Creative is the final task and therefore has no backward-transfer contribution. The right panel shows the aggregate loss-based BWT on a separate scale. Random-B exhibits less loss-based forgetting on Code, Medical, and Math; the Legal difference is negligible and changes sign across seeds. Most of the aggregate difference is concentrated on Medical. Per-seed results are reported in Appendix Q.

produces seed-dependent results: at seed 42 the density-matched gradient arm outperforms the unrestricted one, whereas the ordering reverses at the seven-seed mean. It therefore does not provide a stable explanation of the headline effect.

5.4 Accuracy-Based Validation

To test whether the loss-based forgetting difference is also reflected in task-level predictive performance, we evaluate the same D1 comparison using final accuracy after the five-task sequence. For each seed, the reported quantity is the Grad-B minus Random-B difference in mean option-scoring accuracy over Medical and Legal, the two tasks with discrete-option

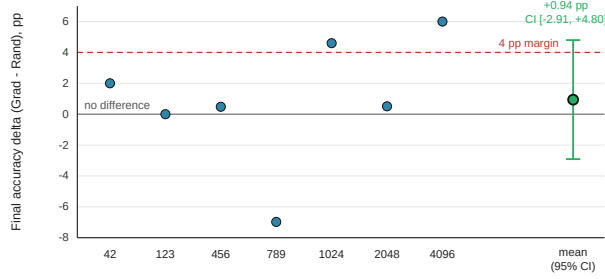


Figure 4: Final-accuracy validation. Blue points are the seven per-seed Grad-B minus Random-B differences in mean final option-scoring accuracy over Medical and Legal after the five-task sequence, in percentage points; green is the seven-seed mean with its 95% interval $[-2.91, +4.80]$ pp. The interval spans zero, so no reliable accuracy difference is detected, and its upper bound exceeds the 4 pp non-inferiority margin, so non-inferiority of Random-B is not established. Per-seed values are in Appendix N.

outputs. It combines learning and retention and is therefore a final-performance check, not an accuracy-based BWT measure.

We detect no statistically reliable accuracy difference between the two masks. The mean Grad-B minus Random-B difference is +0.94 percentage points, with a 95% confidence interval of $[-2.91, +4.80]$ pp. The interval spans zero, and its upper bound exceeds the pre-specified 4 pp non-inferiority margin, so non-inferiority is not established. Thus, the loss-based forgetting difference does not translate into a detectable accuracy difference under this evaluation. Under the observed across-seed variability, a normal-approximation precision calculation indicates that roughly 20 seeds would be needed for a confidence interval narrow enough to resolve a 4 pp margin; this is a planning estimate, not evidence that the current non-inferiority criterion has been met.

Per-seed values and evaluation details are reported in Appendix N. Math is excluded from the accuracy analysis because its generations are truncated under the evaluation setup.

6 Analysis and Extensions

6.1 Sparsity, Scale, and Cross-Architecture Generalization

The headline comparison fixes the sparsity budget at 10%. Sweeping it from 5% to 30% at a single seed, no tested level shows gradient attribution outperforming a shared random mask (Figure 6, Appendix H). The sweep is preliminary and inherits the pool confound rather than isolating the budget, so we report it as a boundary check on the direction of the effect rather than as evidence for the sparsity-constraint view.

Boundaries in scale and task order were probed next, followed by a second architecture. No single-seed probe reverses the conclusion, but a reversed task order shifts BWT by more than any method difference, which is why every conclusion in this paper is scoped to the tested sequence (Table 13, Appendices H–K). On Phi-2, an architecturally distinct family, the within-lora_B direction replicates under the density-unmatched protocol across four seeds, with an effect size comparable to Qwen2.5-3B. Two features of that replication matter for how it should be read: the two Phi-2 designs use different LoRA target sets and are reported separately rather than pooled, and Phi-2 forgets roughly $2.5\times$ less overall, so the magnitude is architecture-dependent even though the direction is not (Appendix K).

6.2 Module Concentration and Calibration Robustness

The gradient mask does not select uniformly across module types: within lora_B it concentrates on output projections and to a lesser degree on up_proj, together accounting

for a little over half of all selections, while down projections receive well below the uniform rate. The pattern is consistent across calibration sources and peaks at middle layers (Appendix J). That concentration invites the reading that it causes the gradient mask’s higher BWT, and the uniform `_B` controller tests it by spreading the same budget across modules. The controller contradicts the reading: three of four seeds land on the gradient side rather than the random side. Because those offsets are taken against different session baselines (Appendix F), we treat the ablation as failing to establish the mechanism rather than as evidence against the concentration itself, and neither the within-lora_`_B` finding nor its density-matched control depends on it (Figure 7). The result is also insensitive to the calibration domain: pooling in-domain rather than out-of-domain calibration changes 42% of the selected parameters, yet the loss-based BWT moves by less than a tenth of the between-seed spread (Appendix L).

7 Related Work

Importance-based CL either regularizes selected parameters (Kirkpatrick et al., 2017; Zenke et al., 2017; Aljundi et al., 2018) or constrains gradient subspaces (Saha et al., 2021; Farajtabar et al., 2020; Wang et al., 2025). Sparse LoRA and pruning likewise assume coordinate identity matters (Yang et al., 2025; Lee et al., 2019; Wang et al., 2020; Sun et al., 2024), while random PEFT subnetworks provide a counterpoint (Xu & Zhang, 2024). We test this assumption within a fixed LoRA pool and budget.

Attribution-Guided CL (Liu et al., 2026) is the closest LLM continual-learning mask, although its published description does not establish all details needed to classify its pool comparison. Non-zero initialization (Meng et al., 2024), adaptive subspaces (Wang et al., 2023; Liang & Li, 2024), gradient projection (Zhao et al., 2024; Choe et al., 2024), and adapter redundancy (Dieudonné et al., 2026) address different settings. Our contribution is a controlled gradient-versus-random comparison within a shared zero-initialized LoRA pool.

8 Conclusion

At equal parameter count within the lora_`_B` pool, the random mask achieves lower loss-based BWT in every seed, an advantage that reproduces in mean and direction in a second hardware/software environment and is not an artifact of density. The density-unmatched within-lora_`_B` comparison points the same way, and the zero-initialization analysis shows why an unrestricted comparison can mix the selection criterion with the parameter pool. On the loss-based evidence, the attribution pass is not justified: removing it eliminates 64 calibration forward-backward steps, an overhead equal to 12.8% of the 500-step training budget (11.3% of the combined 564-step count), and removes the calibration-data requirement. Accuracy remains inconclusive at seven seeds, with the pre-specified non-inferiority criterion unmet. All conclusions are scoped to the tested task order and static masks; a per-task adaptive mask and a better-powered accuracy evaluation are the most informative next tests.

Limitations. The within-lora_`_B` finding is limited to static masks, a single primary task order, and rank-16 LoRA under standard zero initialization. Our primary conclusions are based on loss-based BWT, with accuracy validation limited to the two discrete-option tasks. Absolute BWT also varies across sessions, so all statistical claims rely on within-session paired comparisons (Appendix F).

Reproducibility Statement

Experiments use Python 3.12, PyTorch 2.8.0+cu128 (4-bit NF4 with bf16 compute), Transformers 5.16.1, PEFT 0.20.0, bitsandbytes 0.50.2, datasets 5.0.1, and CUDA 12.8. The primary seven-seed within-lora_B comparison, scale runs, and ablations use a single NVIDIA A800 80GB PCIe GPU. The D1 density-matched control and the M2 uniform-B control ran in a separate session on a Tesla V100-PCIe-32GB under an earlier environment (Python 3.12.3, PyTorch 2.3.0+cu121, Transformers 4.46.3, PEFT 0.13.2, bitsandbytes 0.43.3); as both are within-run paired comparisons, the hardware difference does not affect their paired results. Primary results use Qwen2.5-3B (seven seeds); cross-architecture within-B results use Phi-2 2.78B (four seeds); the confounded Phi-2 comparison uses five seeds under a matched 100-eval-sample protocol (an additional seed-42 observation used 20 eval samples and is reported separately); scale observations use Qwen2.5-0.5B/1.5B/7B (single seed). LoRA is applied to all linear projection layers.

Model training uses AdamW ($\beta_1=0.9$, $\beta_2=0.999$, weight decay 0.01) at learning rate 2×10^{-4} with a linear warmup of 0.03 and cosine annealing over the per-task step budget; LoRA dropout 0.05, rank 16, $\alpha=32$, batch size 1, max length 512, one epoch per task, and top- k 0.1 masking. Five sequential tasks are used: Code (code_alpaca_20k), Medical (medqa, 4-option MC), Legal (lexglue/case_hold), Math (gsm8k), and Creative Writing (Creative_Writing-ShareGPT). Each task uses 500 training and 100 evaluation samples; gradient masks use 64 out-of-domain Alpaca calibration samples. Seeds are 42, 123, 456, 789, 1024, 2048, and 4096; approximately 60 GPU-hours were used in total.

Boundary analyses (rank sweep, task order reversal, scale observations) are single-seed; a method-order check over three seeds is reported in Appendix F. An isolated-process protocol equivalence check confirms that running each method in a separate process reproduces the multi-method BWT to within 9.6×10^{-5} ($10\times$ margin below the pre-specified 10^{-3} threshold). All reported forgetting metrics are loss-based BWT. Code and configuration files are released as anonymized supplementary material (code_anonymous.zip): training and attribution scripts, five-task data preparation, mask generation, the analysis scripts computing BWT and the paired statistics, and pinned dependency manifests for both environments.

Ethics Statement

This work studies parameter masking for continual learning and does not involve human subjects or sensitive data. It does not introduce new capabilities or datasets that raise ethical concerns. On data sources and licensing: the Code task uses CodeAlpaca-20k (CC BY 4.0), the Math task uses GSM8K (MIT), and the Medical and Legal tasks use MedQA and CaseHOLD, which are distributed on Hugging Face with no affirmative license declaration, so we use them for research evaluation only. The Creative task uses a synthetic creative-writing set composed of model-generated outputs rather than user conversations. The Alpaca data used to calibrate the gradient masks is released under CC BY-NC 4.0, a non-commercial license; we therefore make no commercial-use claim for models trained with it. The finding that random selection outperforms gradient attribution within LoRA's B-matrix may reduce computational cost in continual-learning deployments by removing the need for gradient-based parameter attribution. We do not foresee negative societal consequences specific to this contribution.

AI Use Disclosure

AI coding assistants were used for experiment-script development and debugging, and AI assistance aided literature search and manuscript drafting and revision. The research design, experimental methodology, statistical analysis, and scientific conclusions are the authors' own work. All reported results were produced by the described experimental pipeline without AI-generated synthetic data; all analyses and claims were verified by the authors.

References

- Armen Aghajanyan, Sonal Gupta, and Luke Zettlemoyer. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, 2021. URL <https://arxiv.org/abs/2012.13255>.
- Rahaf Aljundi, Francesca Babiloni, Mohamed Elhoseiny, Marcus Rohrbach, and Tinne Tuytelaars. Memory aware synapses: Learning what (not) to forget. In Proceedings of the European Conference on Computer Vision, pp. 139–154, 2018. URL https://openaccess.thecvf.com/content_ECCV_2018/html/Rahaf_Aljundi_Memory_Aware_Synapses_ECCV_2018_paper.html.
- Davis Blalock, Jose Javier Gonzalez Ortiz, Jonathan Frankle, and John Gutttag. What is the state of neural network pruning? In Proceedings of Machine Learning and Systems, 2020. URL <https://arxiv.org/abs/2003.03033>.
- Sang Keun Choe, Hwijeon Ahn, Juhan Bae, Kewen Zhao, Minsoo Kang, Youngseog Chung, Adithya Pratapa, Willie Neiswanger, Emma Strubell, Teruko Mitamura, Jeff G. Schneider, Eduard H. Hovy, Roger B. Grosse, and Eric P. Xing. What is your data worth to GPT? LLM-scale data valuation with influence functions. arXiv preprint arXiv:2405.13954, 2024. doi: 10.48550/arXiv.2405.13954. URL <https://arxiv.org/abs/2405.13954>.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. arXiv preprint arXiv:2305.14314, 2023. URL <https://arxiv.org/abs/2305.14314>.
- Tanguy Dieudonné, Giulia Lanzillotta, Enis Simsar, Louis Barinka, and Thomas Hofmann. When one adapter speaks for many: Discovering low-rank redundancy in continual fine-tuning. arXiv preprint arXiv:2606.28117, 2026.
- Mehrdad Farajtabar, Navid Azizan, Alex Mott, and Ang Li. Orthogonal gradient descent for continual learning. In Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics, volume 108 of Proceedings of Machine Learning Research, pp. 3762–3773, 2020. URL <https://arxiv.org/abs/1910.07104>.
- Jonathan Frankle and Michael Carbin. The lottery ticket hypothesis: Finding sparse, trainable neural networks. In International Conference on Learning Representations, 2019. URL <https://arxiv.org/abs/1803.03635>.
- Robert M. French. Catastrophic forgetting in connectionist networks. Trends in Cognitive Sciences, 3(4):128–135, 1999.
- Soufiane Hayou, Nikhil Ghosh, and Bin Yu. LoRA+: Efficient low rank adaptation of large models. arXiv preprint arXiv:2402.12354, 2024. URL <https://arxiv.org/abs/2402.12354>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. arXiv preprint arXiv:2106.09685, 2021. URL <https://arxiv.org/abs/2106.09685>.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. Proceedings of the National Academy of Sciences, 114(13):3521–3526, 2017. doi: 10.1073/pnas.1611835114. URL <https://arxiv.org/abs/1612.00796>.
- Namhoon Lee, Thalaiyasingam Ajanthan, and Philip H. S. Torr. SNIP: Single-shot network pruning based on connection sensitivity. In International Conference on Learning Representations, 2019. URL <https://arxiv.org/abs/1810.02340>.
- Chunyuan Li, Heerad Farkhoor, Rosanne Liu, and Jason Yosinski. Measuring the intrinsic dimension of objective landscapes. In International Conference on Learning Representations, 2018. URL <https://arxiv.org/abs/1804.08838>.

- Yan-Shuo Liang and Wu-Jun Li. InfLoRA: Interference-free low-rank adaptation for continual learning. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 23638–23647, 2024. URL <https://arxiv.org/abs/2404.00228>.
- Yazheng Liu, Yuxuan Wan, Rui Xu, Xi Zhang, Sihong Xie, and Hui Xiong. Attribution-guided continual learning for large language models. arXiv preprint arXiv:2605.05285, 2026.
- Zhuang Liu, Mingjie Sun, Tinghui Zhou, Gao Huang, and Trevor Darrell. Rethinking the value of network pruning. In International Conference on Learning Representations, 2019. URL <https://arxiv.org/abs/1810.05270>.
- Michael McCloskey and Neal J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In Psychology of Learning and Motivation. Academic Press, 1989.
- Fanxu Meng, Zhaohui Wang, and Muhan Zhang. PiSSA: Principal singular values and singular vectors adaptation of large language models. In Advances in Neural Information Processing Systems, volume 37, pp. 121038–121072, 2024. URL <https://arxiv.org/abs/2404.02948>.
- Jingyang Qiao, Zhizhong Zhang, Xin Tan, Yanyun Qu, Wensheng Zhang, Zhi Han, and Yuan Xie. Gradient projection for continual parameter-efficient tuning. IEEE Transactions on Pattern Analysis and Machine Intelligence, 47(10):9316–9329, 2025.
- Vivek Ramanujan, Mitchell Wortsman, Aniruddha Kembhavi, Ali Farhadi, and Mohammad Rastegari. What’s hidden in a randomly weighted neural network? In 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Ramanujan_Whats_Hidden_in_a_Randomly_Weighted_Neural_Network_CVPR_2020_paper.html.
- Gobinda Saha, Isha Garg, and Kaushik Roy. Gradient projection memory for continual learning. In International Conference on Learning Representations, 2021. URL <https://arxiv.org/abs/2103.09762>.
- Mingjie Sun, Zhuang Liu, Anna Bair, and J. Zico Kolter. A simple and effective pruning approach for large language models. In International Conference on Learning Representations, 2024. URL <https://openreview.net/forum?id=PxoFut3dWW>.
- Qwen Team. Qwen2.5 technical report. arXiv preprint arXiv:2412.15115, 2024. URL <https://arxiv.org/abs/2412.15115>.
- Chaoqi Wang, Guodong Zhang, and Roger Grosse. Picking winning tickets before training by preserving gradient flow. 2020. URL <https://arxiv.org/abs/2002.07376>.
- Chenxu Wang, Yilin Lyu, Zicheng Sun, and Liping Jing. Continual gradient low-rank projection fine-tuning for LLMs. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, 2025. URL <https://aclanthology.org/2025.acl-long.721/>.
- Xiao Wang, Tianze Chen, Qiming Ge, Han Xia, Rong Bao, Rui Zheng, Qi Zhang, Tao Gui, and Xuanjing Huang. O-LoRA: Orthogonal subspace learning for language model continual learning. In Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 10658–10671, 2023. URL <https://arxiv.org/abs/2310.14152>.
- Jing Xu and Jingzhao Zhang. Random masking finds winning tickets for parameter efficient fine-tuning. In Proceedings of the 41st International Conference on Machine Learning, 2024. URL <https://arxiv.org/abs/2405.02596>.
- Shuo Yang, Kun-Peng Ning, Yu-Yang Liu, Jia-Yu Yao, Yong-Hong Tian, Yi-Bing Song, and Li Yuan. Is parameter collision hindering continual learning in LLMs? In Proceedings of the 31st International Conference on Computational Linguistics, pp. 4243–4259, 2025. URL <https://aclanthology.org/2025.coling-main.286/>.

- Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. Language models are Super Mario: Absorbing abilities from homologous models as a free lunch. In Proceedings of the 41st International Conference on Machine Learning, 2024. URL <https://arxiv.org/abs/2311.03099>.
- Friedemann Zenke, Ben Poole, and Surya Ganguli. Continual learning through synaptic intelligence. In Proceedings of the 34th International Conference on Machine Learning, volume 70 of Proceedings of Machine Learning Research, pp. 3987–3995, 2017. URL <https://proceedings.mlr.press/v70/zenke17a.html>.
- Longteng Zhang, Lin Zhang, Shaohuai Shi, Xiaowen Chu, and Bo Li. LoRA-FA: Efficient and effective low rank representation fine-tuning. arXiv preprint arXiv:2308.03303, 2023. URL <https://arxiv.org/abs/2308.03303>.
- Jiawei Zhao, Zhenyu Zhang, Beidi Chen, Zhangyang Wang, Anima Anandkumar, and Yuan-dong Tian. GaLore: Memory-efficient LLM training by gradient low-rank projection. In International Conference on Machine Learning, 2024. URL <https://arxiv.org/abs/2403.03507>.
- Hattie Zhou, Janice Lan, Rosanne Liu, and Jason Yosinski. Deconstructing lottery tickets: Zeros, signs, and the supermask. In Advances in Neural Information Processing Systems, 2019. URL <https://arxiv.org/abs/1905.01067>.

Algorithm 1 Masked continual fine-tuning with a frozen selection mask

Require: base model f_θ with frozen W_0 ; task stream $\{(\mathcal{D}_t)\}_{t=1}^T$; calibration set $\mathcal{D}$; budget k ; rule $\in \{\text{gradient, uniform}\}$
Ensure: frozen mask m ; adapted adapter $W = W_0 + \alpha BA$
1: $\mathcal{S} \leftarrow$ the (A, B) LoRA coordinates $\triangleright$ trainable coordinates
2: if rule = gradient then
3: for each coordinate $i \in \mathcal{S}$ do
4: $g_i \leftarrow \mathbb{E}_{x \sim \mathcal{D}} [\|\partial \mathcal{L}(x) / \partial w_i\|]$ $\triangleright$ Eq. equation 6
5: end for
6: else
7: $g_i \leftarrow \text{Uniform}(0, 1)$ for each $i \in \mathcal{S}$
8: end if
9: $m \leftarrow \mathbb{K}[g_i \geq \tau_k]$ $\triangleright$ top- k rule, Eq. equation 7
10: freeze m $\triangleright$ shared by all tasks; never updated
11: $W \leftarrow W_0$ $\triangleright$ standard zero-init LoRA, $B = 0$
12: for $t = 1$ to T do
13: train the coordinates with $m_i = 1$ on $\mathcal{D}_t$, updating only those
14: evaluate $\mathcal{L}_j^{(t)}$ for all $j \leq t$ $\triangleright$ every task seen so far
15: end for
16: return m , and BWT from Eq. equation 2 over $\mathcal{P} = \{1, \dots, T - 1\}$

A The End-to-End Procedure

The notation is collected in Table 1 in the Methodology. Figure 1(a) shows the same procedure as a diagram; Algorithm 1 states it step by step.

B Attribution Cost

Table 3: Attribution cost on Qwen2.5-3B (BF16). Both columns are from a single full-parameter measurement run (–include-frozen, –full-precision); the exact correspondence and its limits are described below. Peak GPU is `torch.cuda.max_memory_allocated` converted to decimal GB (profiling fields `*_peak_alloc_mb = 13168 / 7102` MiB, i.e. 13.8 / 7.4 GB); Time is the per-task attribution wall-clock recorded by the run (584.9s gradient / 12.1s Wanda).

Method	Peak GPU (GB)	Time (s)	Backward?
Gradient mask	13.8 [†]	584.9	Yes
Wanda proxy	7.4	12.1	No
Shared random / RRM	0 [‡]	0	No
No mask	0 [‡]	0	No

[†]Full-parameter attribution configuration (–include-frozen, scoring all 3,085,938,688 base parameters); no method in this paper uses this configuration. The attribution used in the continual-learning experiments records a ~ 0.73 GB transient allocation above the resident base model during its separate calibration pass (4-bit NF4, seq 512, 64 calibration samples, gradient checkpointing enabled; `overhead_mb = 699.43` MiB). This is not an increase over the ordinary training backward peak; the removable costs are the additional calibration pass, its wall-clock compute, and its data requirement.

[‡]Zero by construction because no separate attribution pass is run, not a measured end-to-end peak-memory saving.

Both measured columns of Table 3 come from a single full-parameter attribution profiling run (–include-frozen, –full-precision), so they are internally consistent; that run scores all 3,085,938,688 base-model parameters, a configuration no reported method uses. We describe the memory figure first, then the time figure.

Gradient attribution requires 13.8 GB peak GPU memory for Qwen2.5-3B (`max_memory_allocated`, 13168 MiB $\rightarrow$ 13.8 GB in decimal units). This figure is from the full-parameter attribution configuration, which scores all 3,085,938,688 parameters

of the base model. The peak comprises four components (the resident base-model weights, the forward and backward activation graph over the full model, the full-parameter gradient accumulator, and the CUDA context), but these were not separately instrumented in the surviving run record, so we report the total rather than a measured decomposition. The backward pass does traverse the base model: attribution computes $\partial\mathcal{L}/\partial w$ over calibration samples, so both the forward and backward graphs span the full parameter space rather than the 29.9M-parameter LoRA surface. This configuration peaks at 13.8 GB, of which $\sim 90\%$ is resident base-model weights (6.23 GB in BF16) plus the full-model gradient buffer (6.23 GB); these are costs the CL configuration does not pay. It is therefore an upper bound on a configuration no method uses, not a measure of attribution’s own overhead. The CL attribution pass records a ~ 0.73 GB transient allocation above the resident base model, but ordinary training also performs backward passes; this is not an incremental end-to-end peak-memory cost. The random baseline removes the separate calibration pass and its time/data costs, not the training backward peak (Table 5).

The 584.9s per-task figure is the wall-clock of that same full-parameter run (`full_alora_time_s = 584.9s`). As a checkpointing-off contrast, an earlier LoRA-only attribution profiling run (4-bit, seq 1024, 256 calibration samples, gradient checkpointing disabled) reports 3,512 MiB of overhead, a different regime (checkpointing off, longer sequences) that should not be combined with the 0.73 GB figure above.

C Pruning Saliency Is Orthogonal to Gradient Importance

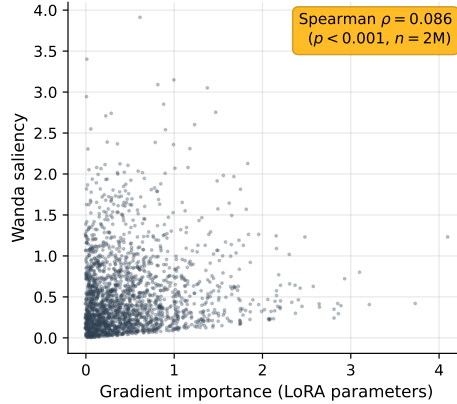


Figure 5: Spearman $\rho = 0.086$ between Wanda and gradient importance (LoRA parameters) on Qwen2.5-3B. These metrics rank parameters nearly independently.

Spearman correlation between Wanda ($|W| \times \text{RMS}(X)$) and gradient importance ($|\partial\mathcal{L}/\partial w|$) across 2M sampled parameters: (Figure 5; $\rho = 0.086$, 95% CI $\approx [0.084, 0.088]$, $n=2M$; correlation-only, measured outside a continual-learning context).

D Confounded Null-Result Statistics

A paired t -test on per-seed BWT (Table 4; random vs. gradient: seeds 42/123/456/789/1024 yield differences $+0.056$, $+0.046$, -0.026 , -0.059 , $+0.028$) gives $t(4) = 0.41$, $p = 0.70$ (two-tailed), Cohen’s $d_z = 0.18$ (small effect). The direction is seed-dependent, consistent with no systematic advantage for either method. Normalizing BWT per task by zero-shot loss ($\Delta L/L_0$) does not change the method ranking. Tables 5 and 6 report different `full_alora` BWT for the same seed because they come from independent runs; the absolute BWT level varies across sessions (§F).

Table 4: Per-seed BWT breakdown (Qwen2.5-3B, rank-16 LoRA, 5 tasks).

Seed	Gradient BWT ↓	Random BWT ↓	No Mask BWT ↓
42	0.491	0.435	0.507
123	0.463	0.417	0.409
456	0.388	0.414	0.539
789	0.357	0.417	0.484
1024	0.486	0.458	0.479
Mean	0.437	0.428	0.484
Std	0.061	0.019	0.047

Table 5: Confounded comparison (Qwen2.5-3B, rank-16 LoRA, 5 tasks, $n=5$ seeds). The last column records whether an additional calibration forward-backward pass is required; its time and data costs are described in Appendix B. Across all tables, bold marks the better value in its column.

Method	BWT ↓	Loss ↓	Calibration pass?
Shared random	0.428 $\pm$ 0.019	1.940 $\pm$ 0.021	No
Gradient mask	0.437 $\pm$ 0.061	1.966 $\pm$ 0.037	Yes
No mask	0.484 $\pm$ 0.047	1.970 $\pm$ 0.033	No

Table 6: Within-lora_B comparison (Qwen2.5-3B, rank-16 LoRA, 5 tasks, 7 seeds). All comparisons are within-run (same method ordering and RNG state). Δ = random_B_only – Grad-Full; negative Δ indicates random-B achieves lower (better) BWT.

Seed	random_B_only BWT ↓	full_alora BWT ↓	Δ
42	0.326	0.491	−0.165
123	0.320	0.413	−0.093
456	0.337	0.356	−0.019
789	0.346	0.364	−0.018
1024	0.336	0.442	−0.106
2048	0.367	0.402	−0.035
4096	0.331	0.398	−0.067
Mean	0.338 $\pm$ 0.015	0.409 $\pm$ 0.046	−0.072

E Within-lora_B Comparison (Density-Unmatched)

The within-lora_B comparison matches the parameter pool (both arms draw from lora_B) but not the density: the gradient mask activates approximately $2\times$ the B parameters of random-B-only. It is therefore an intermediate, density-unmatched step; the density-matched control (D1) is the headline comparison in the main text. All comparisons are within-run (same method ordering and RNG state); Δ = random_B_only – Grad-Full, negative Δ favors random-B. Absolute BWT values carry session-level variation and should not be compared across seeds (§F).

F Variation Across Runs and Method-Order Variation

Three quantities in this paper are sometimes read as competing estimates of one noise level. They are not: each measures a different thing at a different granularity, and the table below separates them.

(a) Same process, protocol split. Running each method in its own process reproduces the multi-method BWT to within 9.6×10^{-5} , an order of magnitude below the pre-specified 10^{-3} threshold. This is a check on the protocol implementation, not on run-to-run variation of the model: two code paths that should agree do agree. The paired t tests instead use the

Table 7: Distinct quantities that are sometimes all called run-to-run noise. The paired tests use the across-seed SD of the within-session contrasts; the protocol-equivalence discrepancy is an implementation check and does not enter inference. Position and cross-session effects are reported as sensitivities rather than as interchangeable error estimates.

Level	What varies		Magnitude	What it measures
(a) protocol-equivalence check	execution only	path	9.6×10^{-5}	numerical agreement between two code paths; not an inferential error term
(b) across-seed paired contrast	independent seed		$SD(\Delta_s) = 0.071$	variability used by the paired significance test
(c) position (ordering) sensitivity	position of the two arms		0.064 (seed 42, gradient_B)	method-position interaction in a three-seed check
(d) cross-session variation	session, hardware, method	hard- and	up to 0.120 (full_alora, seed 42)	method-specific offset of unknown origin

Table 8: Method-ordering sensitivity (D1 density-matched, three seeds). Δ_{pos} is the BWT difference between running at position 1 and position 2. The last two columns express each arm’s positional shift as a fraction of the main D1 contrast ($|\Delta| = 0.155$).

Seed	$ \Delta_{\text{pos}}(\text{rB}) $	$ \Delta_{\text{pos}}(\text{gB}) $	rB / main	gB / main
42	1.5×10^{-3}	6.4×10^{-2}	0.01	0.41
789	3.1×10^{-5}	6.2×10^{-3}	0.00	0.04
2048	8.4×10^{-6}	1.3×10^{-2}	0.00	0.08

across-seed standard deviation of the within-session paired contrasts, $SD(\Delta_s) = 0.071$ for D1.

(b) Same session, arm order swapped. Within one D1 session the two arms run back to back, and reversing their order is a genuine perturbation of the run. At seed 42 the gradient_B arm’s BWT moves by 0.064 between the two orders; at seeds 789 and 2048 both arms stay within 1.4×10^{-2} . This is a within-session positional effect, and the within-run contrast survives it at all three tested seeds. Because this sensitivity analysis covers three rather than seven seeds, it does not establish the population distribution of the position effect.

(c) Cross-session, same seed. The absolute BWT of a fixed method and seed is not reproducible across sessions. For seed 42, full_alora records 0.491, 0.378, and 0.498 in three separate runs (spread 0.120), while random_B_only records 0.326, 0.324, and 0.278 (spread 0.048). The spread is method-dependent, so it does not act as a constant offset that cancels in a comparison.

Method position can shift the absolute BWT of Gradient-B, most notably at seed 42, but reversing the execution order does not reverse the Random-B–Grad-B contrast in any of the three tested seeds. (Table 8)

Why this is not called NF4 non-determinism. Quantization is the same in every run of Table 9, so it cannot by itself explain a cross-session spread while leaving a same-process protocol split at 9.6×10^{-5} . The cross-session spread is method-specific rather than a common shift shared by all arms. We therefore report it as a method-specific cross-session offset of unknown origin and do not attribute it to a mechanism. Two consequences follow. First, only within-run comparisons are used for any claim in this paper. Second, the random_B_only value that enters Table 2 is the D1 session’s own 0.278, taken from the same session as its partner arm, not an average across sessions.

Table 9: Absolute loss-based BWT for the same seed 42 under identical method definitions, across sessions and hardware. Run C is the D1 density-matched session and Run D the uniform-B suppression control (the M2 controller) session, both on a separate GPU and software stack; random_A+B was not run in either. A Run-D entry exists only for methods Run D ran, and it did not train a plain full_alora arm, so that cell is — and its gradient-side arm is gradient_B, entered on its own row. Spread is max – min across the runs in which the method appears.

Method	Run A	Run B	Run C	Run D / spread
Gradient mask (full_alora)	0.491	0.378	0.498	— (0.120)
Gradient-B@10% (gradient_B)	—	—	0.453	0.423 (0.030)
Random mask (random_A+B)	0.435	0.349	—	—
Random B-only (random_B_only)	0.326	0.324	0.278	0.278 (0.048)

Attribution of the level (c) spread. The level (c) spread has no single cause. Method position is not such a cause within a session: set_seed is called before each method’s model load and no random numbers are consumed between iterations, so the seed state entering training does not depend on position. Level (b) shows that position can still matter, and level (c) shows that session identity can. Which of these dominates across a full re-run is not tested here.

Therefore, absolute BWT levels should not be compared across sessions; all statistical claims in the paper use within-session paired contrasts.

G RRM Boundary Analysis

We investigated whether assigning each task a different random mask (Rotating Random Masks, RRM) provides additional benefit beyond shared random masking. RRM generates a deterministic per-task mask via $m_i^{(t)} = \mathbf{1}[\text{hash}(t||i) \bmod 10 < k]$, reducing expected pairwise inter-task parameter overlap from 100% (shared mask) to k^2 (1% at 10% sparsity) (Table 10).

Table 10: RRM boundary analysis (Qwen2.5-3B, single seed per condition except rank-16 reference).

Condition	RRM BWT	Random BWT	Δ	Direction
Rank 4	0.464	0.423	+0.041	Random better
Rank 8	0.427	0.373	+0.055	Random better
Rank 16 (ref.)	0.401	0.435	−0.034	RRM better
Rank 32	0.414	0.390	+0.024	Random better
Reverse order	0.115	0.109	+0.006	Random better

RRM shows a favorable paired difference at rank 16 only. The overlap-reduction hypothesis predicts a consistent RRM advantage; the data do not support this across conditions. RRM’s effect appears condition-dependent rather than robust.

H Sparsity Sweep

We sweep the masking budget over 5%, 10%, 20%, and 30% (single seed) to test whether the within-pool ordering is specific to the 10% sparsity used in the main experiments. Across all four levels, shared random achieves numerically lower BWT than gradient attribution, with no cross-over; the largest margin is at 30%.

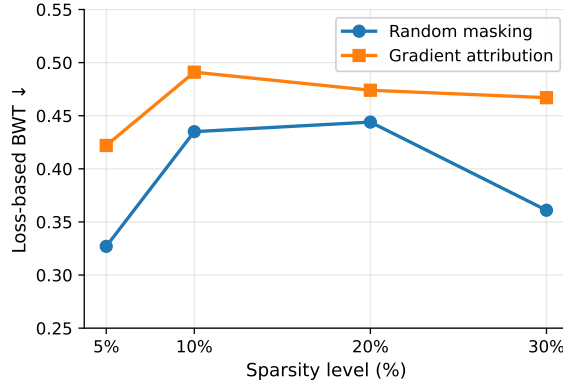


Figure 6: Sparsity sweep (Qwen2.5-3B, single seed). Shared random achieves numerically lower BWT than gradient attribution at all four tested levels.

Table 11: Sparsity sweep numerical values (Qwen2.5-3B, single seed). Random BWT $\leq$ Gradient BWT at all tested levels.

Sparsity	Random BWT ↓	Gradient BWT ↓
5%	0.327	0.422
10%	0.435	0.491
20%	0.444	0.474
30%	0.361	0.467

I Training Stability: Rank-8 No-Mask Divergence

The rank-8 no-mask ablation reveals a discrepancy between two endpoint-based forgetting summaries. Endpoint BWT is 0.176; average forgetting is 0.2837. Training produced a severe loss spike during task 4 (Math): evaluation losses on three prior tasks exceeded 14, compared to values below 4 for all masked conditions at the same rank. This spike subsequently recovered during task 5, yielding deceptively low endpoint BWT. We do not interpret the rank-8 no-mask endpoint BWT of 0.176 as evidence that no-mask outperforms masking.

The same signature recurs at rank 16 and is seed-dependent: at seed 123 the unmasked arm spiked during task 4 with evaluation losses of 16.2/16.0/14.7 on the three prior tasks (code/medical/legal), while every masked arm in that run stayed below 2.5; at seed 42 the unmasked arm did not spike (task-4 evaluations ≤ 1.2). The instability is therefore a property of the unmasked (all-parameter) configuration rather than of a particular rank, and it appears in a minority of seeds.

J Mask Concentration by Module

The `o_proj` suppression ablation and module-concentration figures, discussed in §6, are reproduced below together with the module-concentration table for reference.

K Generalization and Boundary Checks

Scale Observations We additionally observe the within-pool comparison at three parameter scales (0.5B, 1.5B, 7B; single seed each). These are observational rather than multi-seed, and at 0.5B and 1.5B only the shared-random and no-mask conditions were run, so no gradient attribution baseline is available for a direct comparison there.

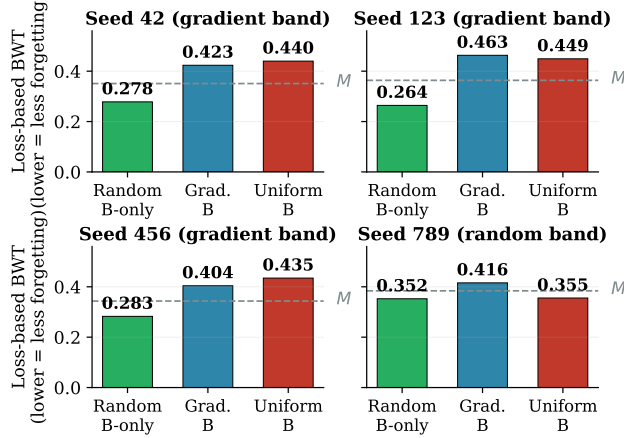


Figure 7: The suppression control’s outcome is seed-dependent. All three arms train 1,651,508 parameters, which is 5.5% of the full LoRA pool (each arm draws 10% of the `lora_B` sub-pool, and `B` holds 55% of the pool’s parameters), so this is the density-matched setting throughout. The uniform-B controller redistributes the gradient mask’s budget uniformly across LoRA modules. At seed 42 it sits in the gradient band (0.440 above gradient-B’s 0.423 and well above random-B-only’s 0.278); seeds 123 and 456 also land on the gradient side; at seed 789 it falls to the random band (0.355 below gradient-B’s 0.416, alongside random-B-only’s 0.352). Three of four seeds fall on the gradient side. M2 is an independent session from D1, so the same-seed gradient-B values differ between them (e.g. seed 42: 0.423 here vs 0.453 in the D1 table; seed 789: 0.416 vs 0.387; §F). We therefore do not treat this ablation as establishing the `o_proj`-concentration mechanism (§6).

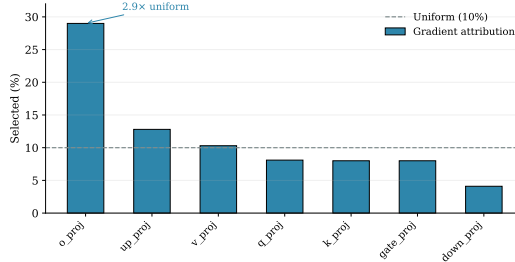


Figure 8: Gradient-mask concentration by LoRA module (Qwen2.5-3B). Gradient attribution selects 29% of `o_proj` parameters (2.9× the 10% uniform rate) and is under-represented in `down_proj` (4.1%); the dashed line marks uniform 10% selection.

Task Ordering (single seed) A reversed-order experiment yields BWT ≈ 0.11 for all masking methods, compared to ≈ 0.42 under the standard order, a nearly 4× reduction that far exceeds the method differences measured within this experiment (within-session method contrasts of order 10^{-2}). This is a single seed, so the 4× magnitude is a qualitative pointer, not a quantitative estimate; it is reported to show that task ordering dominates the forgetting landscape, not as a measured effect size. The conclusions of this paper apply only to the tested Code→Medical→Legal→Math→Creative sequence; additional task orderings with multi-seed replication are needed before the null result can be generalized.

Cross-Architecture Comparison Both cross-architecture designs are tabulated in Table 14: the confounded design (gradient selecting B-only, random selecting A+B) and the within-`lora_B` design (both methods restricted to `lora_B`). The two designs are not comparable row-for-row because the Phi-2 within-B rows use a different LoRA target set; §6.1 discusses both.

Table 12: Gradient mask concentration by LoRA module (Qwen2.5-3B). Random masking distributes uniformly at 10% per module. Gradient attribution selects exclusively lora_B parameters (all lora_A receive zero importance due to $B = 0$ initialization).

Module	Gradient density	Random density	Ratio
o_proj	29.0%	10.0%	$2.9\times$
up_proj	12.8%	10.0%	$1.3\times$
v_proj	10.3%	10.0%	$1.0\times$
q_proj	8.1%	10.0%	$0.8\times$
k_proj	8.0%	10.0%	$0.8\times$
gate_proj	8.0%	10.0%	$0.8\times$
down_proj	4.1%	10.0%	$0.4\times$

Table 13: BWT at additional scales (single seed, observational).

Scale	Random BWT $\downarrow$	Gradient BWT $\downarrow$	No Mask BWT $\downarrow$
0.5B	0.343	—	0.431
1.5B	0.309	—	0.356
7B	0.447	0.441	0.514

Table 14: Cross-architecture comparison on Qwen2.5-3B and Phi-2 (2.78B, GPT-NeoX-style), with both arms within a row sharing one target set. Qwen2.5-3B targets q/k/v/o/gate/up/down_proj (pool 29,933,568), whereas Phi-2 uses GPT-NeoX-style q,k,v,fc1,fc2,dense projections (pool 23,592,960) in the confounded design. In the within-lora_B design, the Phi-2 rows omit dense (pool 20,971,520, 12.5% smaller), so the two Phi-2 designs are not comparable row-for-row. The within-lora_B rows are matched on pool but not density (the gradient mask activates approximately $2\times$ the B parameters of Random-B), as on Qwen2.5-3B.

Architecture	Comparison	Grad. BWT $\downarrow$	Rand. BWT $\downarrow$	Δ
Qwen2.5-3B ($n=5$)	Confounded	0.437 ± 0.061	0.428 ± 0.019	-0.009
Phi-2 ($n=5$)	Confounded	0.158 ± 0.019	0.183 ± 0.016	$+0.025$
Qwen2.5-3B ($n=7$)	Within-B	0.409 ± 0.046	0.338 ± 0.015	-0.072
Phi-2 ($n=4$)	Within-B	0.149 ± 0.036	0.065 ± 0.017	-0.084

L In-Domain Calibration Ablation

The primary experiments use out-of-domain Alpaca calibration data (64 samples). To test whether this choice drives the null result, we repeat the gradient-mask experiment with pooled in-domain calibration: 60 samples drawn equally from the five CL tasks.¹ All other parameters are identical (Qwen2.5-3B, seed 42, rank 16, 10% sparsity).

Loss-based BWT is 0.495 vs 0.491 with Alpaca calibration ($\Delta = 0.004$, 0.8% relative), well within the 5-seed standard deviation of 0.061. Random and no-mask controls are numerically identical across both runs, confirming infrastructure determinism. The cross-calibration overlap (Alpaca vs pooled) is Jaccard = 0.585: changing the calibration domain alters 42% of selected parameters. Despite this substantial mask difference, BWT changes by only 0.004. This robustness is of the outcome, not the mask: the mask does change, but among gradient-derived masks the specific 10% selected has little influence on forgetting, whereas swapping the selection criterion entirely (gradient for random) does, as the density-matched control shows (§5.1).

¹ $\lfloor 64/5 \rfloor \times 5 = 60$; a 6.25% size difference from integer division.

M B=0 Restriction Asymmetry

The within-lora_B comparison varies the selection criterion while holding the parameter pool fixed. That the pool restriction is the right control to hold follows from an asymmetry between the two arms at seed 42 (Table 15). Restricting the gradient mask to lora_B changes nothing, because under $B = 0$ every lora_A importance score is already zero and the restriction is a no-op. The same restriction applied to the random arm removes the lora_A dilution and improves BWT by 0.025. A restriction that does nothing to one arm and helps the other is evidence that the two arms were drawing from different pools before the restriction was applied, which is what makes the pool-matched comparison the informative one.

Table 15: Effect of restricting the mask to lora_B (seed 42). For gradient attribution, the restriction is a no-op by construction ($B=0$ collapses every lora_A importance score to zero); for random, restricting to B removes lora_A dilution and improves BWT by 0.025. This asymmetry demonstrates that the within-B comparison varies the selection criterion, not the parameter pool.

Method	Unrestricted BWT	B-only BWT	Gap
Gradient	0.378	0.378	+0.000
Random	0.349	0.324	−0.025

N Per-Seed Accuracy and Matched Loss Contrast

The accuracy evaluation is reported in the main text, §5.4, together with the option-scoring protocol; its per-seed values and the matched loss contrast are tabulated below. (Table 16)

O The Math Task: Truncation Audit

Loss on math decreases with training while accuracy on the same task collapses to zero for all arms: peak accuracy at math’s own training phase is 0.04 and every arm ends at 0.00. The numeric-match function accepts a bare number, a number with surrounding text, and a complete solution, and the prompt prefixes are identical; nevertheless, the interaction between output format and the 256-token limit makes the generation-based score unreliable. A generation audit reports 56–95% of generations across all tasks and arms reaching the cap without an end-of-sequence token, and for math specifically 81–90% of generations reach the cap. The last number in a truncated solution is often an intermediate quantity, so the reference answer is absent from the value the scorer inspects. Every arm shares the same cap and scorer, so the bias is common rather than arm-specific. We do not treat the math column as a forgetting result: under option-based scoring math yields no score, and under generation scoring it is confounded with truncation.

P Positioning Against Gradient-Attribution Families

Table 17 places our contribution against the gradient-attribution families we are positioned against, on the two axes that drive the study.

Table 18 lists the selection rules this paper is positioned against and marks which we train under one protocol. Only the rules we train contribute empirical numbers here; the rest are compared by their published design, with none of their reported numbers presented as a measurement of ours.

Q D1 Per-Task and Per-Seed Decomposition

Table 19 gives the seven-seed mean per-task contribution. Table 20 gives the same quantity for each seed, so the task-level pattern can be checked against the seed-level one; the main

Table 16: Col. 2 is the Grad-B minus Random-B difference in mean final option-scoring accuracy over Medical and Legal after the five-task sequence, in percentage points; it is a final-performance measure, not accuracy-BWT. Positive values favour Grad-B. Col. 3 is the loss-based BWT difference on the same A800 environment, where positive values indicate greater forgetting for Grad-B. Col. 2 carries the 95% t -interval on the seven-seed mean. Col. 3 has mean $+0.152$ (SD 0.0749 , $t = 5.37$, CI $[+0.083, +0.221]$, 7/7 same direction). The V100 contrast is $+0.155$ under the same [Grad – Random] convention. Across matched seeds, the A800-minus-V100 contrast difference has mean -0.0033 and SD 0.0441 , so replication is strongest at the mean and directional levels rather than as seed-wise numerical identity.

Seed	Final accuracy Δ [Grad – Rand], pp	Loss BWT Δ [Grad – Rand]
42	+2.00	+0.2320
123	0.00	+0.1647
456	+0.48	+0.1300
789	−6.98	+0.0786
1024	+4.60	+0.0514
2048	+0.51	+0.2572
4096	+6.00	+0.1492
Summary (column 2 in percentage points, column 3 in BWT units)		
Mean	+0.94 pp	+0.152
SD	4.17 pp	0.0749
95% CI	$[-2.91, +4.80]$ pp	$[+0.083, +0.221]$

Table 17: Positioning against gradient-attribution families. "B=0 pool" isolates the lora_B pool; "within-pool grad-vs-random" compares gradient to a matched random baseline in a shared pool.

Method	B=0 pool	Within-pool grad-vs-random
EWC / SI / MAS	No	No
PiSSA ($B \neq 0$)	N/A by init	No
Attribution-Guided CL	Unclear [†]	Unclear [†]
Ours (within-lora_B)	Yes	Yes

[†]The published description does not establish all initialization, attribution-timing, baseline-pool, and mask-scope details required to classify these two axes; we therefore do not infer a pool mismatch.

text reports the same decomposition as a figure (Figure 3). Each entry is $\text{loss_matrix}[4][j] - \text{loss_matrix}[j][j]$ for prior task j ; the final column is the mean over the four tasks, which is the per-seed aggregate Δ of Table 2.

R Ruling Out a Learning Deficit in the D1 Gap

The D1 density-matched comparison (both arms 10% of lora_B, 1,651,508 parameters) shows random-B-only achieving lower BWT than gradient-B. One rival reading is that gradient-B’s top-10% is the more concentrated half of the original 20% mask, so gradient-B learns each task less well and the gap is a learning deficit rather than a retention difference. The per-task diagonal losses (each task’s loss immediately after it is learned) exclude this: gradient-B’s diagonal loss is lower than random-B-only’s in all 35 seed $\times$ task cells (7 seeds $\times$ 5 tasks), with no exceptions.

A second rival reading is that BWT, being a difference of losses, is inflated for gradient-B simply because it starts from a lower per-task loss. Reconstructing each seed’s mean per-task endpoint (BWT plus the mean diagonal over the first four tasks) closes this from both sides: gradient-B starts lower yet ends higher in 6 of 7 seeds, so its larger BWT reflects greater forgetting rather than a lower baseline (Table 22).

Both checks are loss-based (accuracy is measured separately in the accuracy branch, §5.4), use a single task order, and cover Qwen2.5-3B only. The endpoint reconstruction is a

Table 18: Selection rules this paper is positioned against. The last column marks the four rules we train and evaluate under one protocol (§3.3); the rest are compared by design only, and none of their published numbers is cited as a result of ours.

Rule	What it scores, and what for	Run
EWC (Kirkpatrick et al., 2017); MAS (Aljundi et al., 2018)	Fisher information, output sensitivity; a penalty, not a mask	no
SNIP (Lee et al., 2019); GraSP (Wang et al., 2020)	gradient at initialisation; one-shot pruning mask	no
Wanda (Sun et al., 2024)	weight $\times$ activation magnitude; one-shot pruning mask	no
LoRA-FA (Zhang et al., 2023)	none, structural; freezes lora_A, trains lora_B	no
Grad-Full, Grad-B (§3.3)	calibration gradient magnitude; static CL mask, full pool (Grad-Full) or lora_B at 10% (Grad-B)	yes
Random-B; Random A+B (Xu & Zhang, 2024)	none, uniform; static CL mask, lora_B at 10% (Random-B) or full pool (A+B)	yes

Table 19: D1 per-task decomposition of loss-based BWT, averaged over the seven seeds. Each entry is $\text{loss_matrix}[4][j] - \text{loss_matrix}[j][j]$ for prior task j , that task’s contribution to the aggregate; the last column is its share of the aggregate Δ in absolute value. Creative is the final task and contributes no forgetting term. The aggregate is concentrated on medical (79.9%) but not confined there: excluding medical the matched contrast remains -0.042 (7/7 negative), while code and math are each negative in 7/7 seeds and legal is not established.

Prior task	random-B	gradient-B	Δ	share of Δ
Code	-0.021	0.025	-0.046	7.4%
Medical	0.998	1.493	-0.496	79.9%
Legal	0.007	0.010	-0.003	0.5%
Math	0.191	0.266	-0.076	12.2%
Aggregate	0.294	0.449	-0.155	100%

Table 20: Per-seed, per-task D1 decomposition ($\Delta = \text{random-B} - \text{gradient-B}$ for each prior task). Negative favours random-B. Legal is the only task whose sign is not consistent across seeds.

Seed	Code	Medical	Legal	Math	Aggregate
42	-0.040	-0.409	-0.002	-0.249	-0.175
123	-0.002	-0.698	-0.002	-0.052	-0.188
456	-0.074	-0.381	-0.004	-0.033	-0.123
789	-0.063	-0.054	+0.001	-0.023	-0.035
1024	-0.044	-0.410	-0.011	-0.021	-0.121
2048	-0.041	-0.950	+0.010	-0.062	-0.261
4096	-0.059	-0.568	-0.014	-0.091	-0.183
Mean	-0.046	-0.496	-0.003	-0.076	-0.155

derived quantity whose per-seed differences are of the same magnitude as the cross-run drift documented in Appendix F; it is therefore offered as a qualitative directional exclusion, not a significance claim.

Table 21: Diagonal loss (mean $\pm$ std over the seven D1 seeds), by task. Lower is better. gradient-B is lower than random-B-only in every cell (Table 21); Grad-Full is the uncontrolled reference. Loss diagonals only; the D1 run records contain no accuracy matrix (accuracy is measured separately in the accuracy branch, §5.4).

Task	random_B_only	gradient_B	full_alora
code	0.952 ± 0.111	0.901 ± 0.100	0.898 ± 0.099
medical	1.190 ± 0.040	1.152 ± 0.040	1.144 ± 0.035
legal	0.388 ± 0.035	0.368 ± 0.038	0.365 ± 0.037
math	3.838 ± 0.103	3.652 ± 0.096	3.642 ± 0.100
creative	2.104 ± 0.053	2.067 ± 0.053	2.056 ± 0.054

Table 22: Derived per-task endpoint loss (BWT + mean diagonal over the first four tasks), per seed. gradient-B’s endpoint is higher in 6 of 7 seeds; seed 789 is the exception. These are derived quantities of the same order as cross-run drift, reported for the qualitative direction only.

Seed	rB start	rB end	gB start	gB end (diff)
42	1.543	1.821	1.478	1.931 (+0.110)
123	1.632	1.896	1.553	2.006 (+0.110)
456	1.575	1.858	1.494	1.900 (+0.042)
789	1.548	1.900	1.475	1.862 (−0.038)
1024	1.566	1.918	1.489	1.962 (+0.045)
2048	1.665	1.896	1.587	2.078 (+0.182)
4096	1.618	1.913	1.553	2.031 (+0.118)